

Supplier Blindspot

Which suppliers are quietly costing Arora Traders money, how much, and what to say to them. This document explains how the answer was found, in plain words, and what was assumed along the way.

Live app [supplier-intelligence-iota.vercel.app](#) Source code [github.com/Prathamnm/supplier-intelligence](#)

THE ANSWER IN ONE MINUTE

- Three suppliers are the problem: Tata Steel Service, Thakur Steel, Nutan Tubes.** They get only 9% of the orders but cause 54% of all the money lost, **₹10.46 Cr over 3 years**.
- The money is lost in plain ways:** material that was billed but never arrived, material customers sent back, and material rejected at Arora’s own gate. Every rupee is worked out order by order and can be checked.
- 140 of the 380 customer returns had no supplier written down.** A model traces each one to its most likely supplier. When tested on returns where the answer was known, it put only 5.3% of the value on the wrong supplier, against 48% for the rule suggested in the assignment.
- The result was double-checked.** The dataset secretly marks which suppliers are bad. That column was locked away and never used; opened at the end, it named the same 3 of 3 suppliers.
- Each bad supplier gets a ready-to-use negotiation brief,** with the money they owe, the orders that prove it and what they are likely to argue back. The recommendation is to replace them, and existing suppliers who can take over each material are named.

LOST TO THE BOTTOM 3

₹10.46 Cr

over 3 years, from 9% of the orders

LOST ACROSS ALL SUPPLIERS

₹19.51 Cr

34 suppliers, 4,800 orders

UNTRACED RETURNS

140 of 380

traced to their likely supplier

DOUBLE-CHECK

3 of 3

match the dataset’s hidden answer

1 The problem, in plain words

Arora Traders buys steel from 34 suppliers. Some deliver less than they bill for, some deliver late, and some deliver material that fails inspection or comes back from customers. Nobody had added this up per supplier, and 37% of the customer returns don’t even say which supplier the material came from. That is the “blindspot”.

The data covers 2021-01-01 to 2024-01-01: 4,800 purchase orders, their deliveries and payments, 380 customer returns, a market price list and a supplier list. The assignment asked for five things, and each has a clear home:

WHAT WAS ASKED	MARKS	WHERE TO FIND IT
Money lost per supplier , in rupees	40%	Section 3; every supplier’s page in the app
Trace returns that have no supplier recorded	35%	Section 4; the app’s “Untraced returns” page
Negotiation briefs for the worst suppliers	25%	Section 7; the app’s “Briefs” page (printable PDFs)
Clean the data; build a scorecard ; recommend who to replace	—	Sections 2, 5 and 7; the app’s Overview

2 How it works, in five steps

One command (`python -m pipeline.run`) takes the six raw files and produces everything: the scorecard, the rupee figures, the traced returns, the briefs, the website's data and this document. The same steps run in the cloud when someone uploads their own files on the website.

STEP 1

Read and clean

Recognise each file by its columns, fix Indian number and date formats, and stop loudly if anything doesn't add up.

STEP 2

Trace the returns

Work out the most likely supplier for each return with no supplier recorded.

STEP 3

Count the money

Put a rupee figure on every short delivery, return and rejection, order by order.

STEP 4

Score and rank

Score each supplier out of 100 on four checks and rank all 34.

STEP 5

Write the briefs

Turn the evidence into a two-page brief for each of the worst suppliers.

What cleaning the data turned up

Before any scoring, every assumption the arithmetic depends on was tested. A few findings shaped the whole approach:

- **The answer was hidden in the data.** The supplier list has a column marking the bad suppliers. Using it would be cheating, so it is removed the moment the files are read and only opened at the very end, as a check (section 6).
- **Suppliers bill for what was ordered, not what arrived.** This is true on every order, so the gap between what was billed and what arrived is money paid for nothing.
- **The market price list can't judge suppliers.** It covers only 6 of the 18 materials, and everyone pays well above it. Prices are therefore compared with what *other suppliers* charged for the same material in the same quarter.
- **Some fields contradict each other**, for example "days late" in payments doesn't match its own dates. These are recalculated from the raw dates, and every fix is logged (23 findings in total).

3 Counting the money lost - 40% of marks

Money is counted per purchase order and then added up per supplier. The assignment's own formula is used exactly as given:

Money lost = short delivery + customer returns

Short delivery = what was billed – (what actually arrived × the agreed price)

Customer returns = tonnes returned × the supplier's usual price for that material

For a return with no supplier recorded, the value is shared out by the model's probabilities (section 4), so every returned tonne is counted exactly once.

Two further losses are real but outside that formula, so they are **shown separately** and never mixed into the assignment's figure: **material rejected at Arora's gate** (rejected tonnes × price) and the **cost of handling a return** (an assumed 15% of its value).

Worked example: Tata Steel Service

WHERE THE MONEY WENT	AMOUNT	HOW IT WAS WORKED OUT
Billed but never delivered	₹1,04,34,509	124.7 tonnes short across 141 orders
Returned by customers	₹1,37,02,499	39 returns on record, plus about 25 untraced returns that are most likely theirs
The assignment's formula	₹2,41,37,008	
Rejected at our gate	₹73,60,582	88.5 tonnes failed inspection
Handling the returns	₹20,55,375	15% of return value (assumed)
Total	₹3,35,52,964	11.2% of everything bought from them

The worst suppliers against everyone else

Share of orders		9%
Share of money lost		54%

SUPPLIER	RANK	SHORT DELIVERY	RETURNS	FORMULA TOTAL	EVERYTHING
Tata Steel Service	34 of 34	₹1.04 Cr	₹1.37 Cr	₹2.41 Cr	₹3.36 Cr
Thakur Steel	33 of 34	₹1.06 Cr	₹1.37 Cr	₹2.43 Cr	₹3.37 Cr
Nutan Tubes	32 of 34	₹1.19 Cr	₹1.49 Cr	₹2.69 Cr	₹3.73 Cr
Modern Metals	31 of 34	₹13.80 L	₹29.21 L	₹43.01 L	₹47.39 L
Garg Enterprises	30 of 34	₹14.23 L	₹22.88 L	₹37.11 L	₹40.55 L
Sai Steel	29 of 34	₹13.32 L	₹23.80 L	₹37.13 L	₹40.70 L
All 34 suppliers				₹15.99 Cr	₹19.51 Cr

Does it match the assignment's own estimate? The assignment guesses the three bad suppliers cost **₹8–12 lakh a year** on ₹70–85 lakh of monthly buying. The data actually shows far more buying (₹27.26 Cr a month), so the raw totals are bigger. Scaled back to the assignment's buying level, the worst suppliers lose **₹7.95 L – ₹9.65 L a year** more than an average supplier would, which is inside the assignment's range. It was reached independently, not tuned to fit.

4 Tracing returns with no supplier · 35% of marks

140 customer returns say what came back, when and why, but not which supplier sent it. They can't simply be ignored, because that would let the worst suppliers off the hook for a large share of their returns.

The idea

Think of it like narrowing down a suspect list. For each untraced return, the **suspects** are the suppliers who had delivered that material before the return date. For each suspect, the model looks at clues that were known on that date, never later:

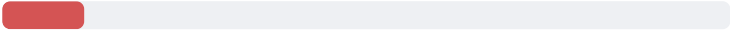
- How recently they delivered this material
- How much of this material they supplied recently
- How much of their material we rejected

- How often they delivered short
- How late they usually deliver
- How many returns were already traced to them

It learns how much each clue matters from the 240 returns where the supplier is recorded, then gives every suspect a probability. The return's value is shared out by those probabilities rather than pinned on one supplier.

Does it work? Tested with the answer hidden

The model was tested on the 240 returns whose supplier is known: the answer was hidden, the model guessed, and the guesses were compared with the truth (5-fold cross-validation). The key question for the money is **how much of the returned value ends up on the wrong supplier**. Lower is better:

This model		5.3%
Blame whoever delivered that material last (the assignment's rule)		47.9%
Split by who supplied most of that material		47.1%
Split equally between all suppliers		47.0%

So the model gets about **95% of the value right**, while every simple rule, including the one the assignment suggested, gets roughly half wrong. That rule ("blame whoever delivered that material most recently") turned out not to match the returns where the answer is known.

Be honest about the limits. For a single return the model often can't be sure, because several suppliers could have sent it and nothing on the return tells them apart. It picks the right supplier first 21% of the time and has it in its top three 56% of the time. That is why traced returns count towards the totals as an estimate, but **a supplier is only ever asked to pay for returns that are on record**.

5 The scorecard

Every supplier gets a score out of 100 (higher is better), built from four checks:

CHECK	WEIGHT	WHAT IT MEASURES
Short delivery	30%	How much of what was ordered never arrived
Quality	25%	How much was rejected at our gate or returned by customers
Late delivery	25%	How many days late orders arrive, on average
Price	20%	How much more they charge than other suppliers for the same material

A few rules keep the scores fair:

- **Compared with a typical supplier, not the worst.** Matching the typical supplier earns 100; only being clearly and meaningfully worse loses points. If nobody is bad, nobody is flagged. This was tested on data where every supplier was fine, and nobody was flagged.
- **Small suppliers aren't punished for bad luck.** A supplier with only a handful of orders is pulled towards the average until there's enough evidence, so two bad orders can't put them at the bottom.
- **Price needs proof.** Prices vary a lot from order to order, so a supplier is only charged for high prices if a statistical test says it isn't chance. No supplier passed, so ₹0 is claimed for price. The briefs still ask for a price match where a supplier charges above the others.

- **Relationship length is ignored.** How long someone has supplied Arora says nothing about how well they deliver, so it isn't used. A test proves changing it changes no score.
- **The weights don't decide the answer.** The same suppliers come out at the bottom under all 5 different weightings tried, including equal weights.

6 The double-check

With the ranking finished, the hidden "bad supplier" column was opened for the first time. It marks VS03, VS12, VS19. The analysis, which never saw it, had placed exactly these suppliers at the bottom: ranks VS03 #34, VS12 #32, VS19 #33 of 34, **69 points clear** of the next supplier. That is a large, unambiguous gap, not a near miss.

7 Negotiation briefs and who to replace · 25% of marks

Each of the 5 worst suppliers gets a two-page A4 brief a buyer can take into a meeting:

- **Page one:** the rupee figure and how it was worked out, three or four numbered asks with the evidence for each, what the supplier is likely to argue back, and when to walk away.
- **Page two:** the purchase orders to put on the table, and a breakdown by material.

Every ask quotes a figure the supplier can check against their own invoices. For example, to Tata Steel Service:

"Issue a credit note of ₹1,04,34,509 for material billed but not delivered, and invoice on GRN-received quantity from the next order."

Evidence: 141 of 141 orders arrived short — 124.7 MT of 3,541.4 MT ordered (3.5% vs 0.7% for the rest of our panel).

Recommendation: replace these suppliers

REPLACE	MAIN PROBLEMS	COST	WHO CAN TAKE OVER THEIR BIGGEST MATERIALS
Tata Steel Service	Short deliveries Late deliveries Poor quality	₹3.36 Cr	MS Flats → Ratan Steel, Prakash Industries; MS Rods → MSP Steel, Laxmi Hardware; SS Pipes → Viraj Profiles, Shyam Metalics
Thakur Steel	Short deliveries Late deliveries Poor quality	₹3.37 Cr	CR Sheets → Global Metals, Bhushan Steel; GI Wire → Precision Steel, Prakash Industries; ERW Pipes → Bhavani Metals, Om Metals
Nutan Tubes	Short deliveries Late deliveries Poor quality	₹3.73 Cr	HR Coils → Viraj Profiles, Western Steel; Spring Steel → SAIL Distribution, Bhushan Steel; MS Rods → MSP Steel, Laxmi Hardware

The alternatives are existing suppliers who already deliver that material well, so switching doesn't mean finding someone new.

8 Assumptions and limitations

- **Value of a return:** the supplier's usual price for that material. **Handling a return:** assumed to cost 15% of its value, and kept outside the assignment's formula.
- **Late-delivery penalty** in the briefs (0.5% a day, capped at 5%) is a negotiating position, not a loss, and is never added to any total.
- **Arora pays late too.** This is worked out from invoice dates and shown in each brief as the supplier's likely comeback. It doesn't change their score.
- **Untraced returns are assumed to be like the traced ones.** If clerks tended to record some suppliers' returns more than others', the model would lean the wrong way, and the data can't reveal that. This is the main reason only recorded returns are ever claimed. The lasting fix is simple: write a batch number on every return.

- **Nothing is tuned to this dataset.** Suppliers, materials and thresholds are discovered from the data, and every analysis, including each upload, is worked out from its own files alone.

9 How it was checked

An independent recount. A separate script recalculates every headline figure straight from the raw files, without using any of the analysis code, and confirms the website shows the same numbers.

Test data with a known answer. Nine made-up datasets with planted bad suppliers: a 120-supplier panel, suppliers who turn bad only in the final year, a small but genuinely bad supplier, messy Indian spreadsheet exports, and one where nobody is bad. In all nine, the right suppliers were found, for the right reason, with no false alarms.

Automatic tests check the money formulas, that the hidden answer never reaches the scoring, that relationship length has no effect, the upload service, and every page of the website in a real browser.

Try it. The live website opens on these results. On its Upload page, anyone can add their own six files (any file names, Excel or Tally exports as they are) and get the same analysis for their suppliers.